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Bayesian Local Sensitivity to Non-ignorability with General Missingness of Outcomes and Covariates

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Abstract:

A major concern in statistical analysis is missing data. Analysis frequently operates under the presumption that missingness is ignorable when it occurs in the study. It is critical to evaluate the effect of missingness on the most important Bayesian inferences because ignorability is an untestable assumption in the absence of data augmentation. Previously, in a Bayesian framework, the influence of missing outcomes' non-ignorability was studied using local sensitivity indices. In this paper, for the first time, the Bayesian index of local sensitivity to non-ignorability for data exposed to general missingness in outcome and covariates will be extended. My results indicate that the Bayesian estimates of parameters of interest via local sensitivity approximation are more accurate than complete-case analysis and ignorable model. I will evaluate the features of my proposed indices through several simulated studies.

Keywords: Local Sensitivity to Non-ignorability, General Missingness, Bayesian Approach, and Local Sensitivity Indices.

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1 Introduction

The selection model pertaining to incomplete data articulates the joint distribution of the hypothetical complete data alongside the indicators of missingness as the marginal distribution of the complete data multiplied by the distribution of the missingness indicators conditioned on the complete data. The latter component is referred to as the missingness mechanism. [Rubin \(1976\)](#) clarified the conditions under which it is permissible to disregard the stochastic characteristics of the missingness mechanism without jeopardizing the integrity of model-based inferences: Missing at random (MAR) posits that the occurrence of missingness is independent of the values of the missing items, whereas parameter distinctness (PD) contends that the parameters that dictate the distribution of the hypothetical complete data and those that govern the missingness mechanism reside within a factorizable parameter space (for likelihood inference) or are a priori independent (for Bayesian inference). The conjunction of MAR and PD ensures the ignorability of the missingness mechanism in both likelihood-based and Bayesian inferential frameworks. Analyses that disregard the intricacies of the missing data mechanism consequently presuppose, either implicitly or explicitly, an MAR framework alongside a PD Parameterization ([Little , 1992](#); [Little and Rubin , 2002](#); [Paik and Sacco , 2000](#); [Satten and Carroll , 2000](#)). A significant concern is that the presumption of MAR is often not tenable across numerous practical applications; furthermore, it is not feasible to evaluate this assumption in a robust manner by solely scrutinizing the available data.

Typical selection models incorporate a nonignorability parameter that dictates the dependence of the probability of missingness on the values of the notional underlying data. It is customary to parameterize such models in a manner that renders the missingness mechanism MAR when the nonignorability parameter is assigned a value of 0. Estimating such a parameter is rarely feasible without the imposition of strong assumptions, and even under such circumstances, the resulting inferences may exhibit nonrobustness and pose computational challenges ([Diggle and Kenward , 1994](#); [Robins et al. , 2000](#); [Scharfstein et al. , 2003](#)). Sensitivity analysis endeavors to mitigate this issue by deriving estimates of the other parameters across a spectrum of presumed values for the nonignorability parameter; should the estimates exhibit robustness to variations in its value, one concludes that the results are insensitive. A local sensitivity analysis investigates the impact on inferences of small departures from MAR assumption ([Troxel et al. , 2004](#); [Zhang and Heitjan , 2007](#); [Eftekhari Mahabadi , 2018](#); [Momeni Roochi and Eftekhari Mahabadi , 2024](#)).

Most of these analyses have primarily addressed scenarios wherein the missing data is confined to outcome variables and postulate a singular scalar non-ignorability parameter, thus facilitating a more straightforward interpretation. [Gao et al. \(2016\)](#) considered the missingness in both outcome and covariate simultaneously, but in the real world of data missingness is often in a general form, that is in some cases response variable is missed but the covariate is observed and vice versa. To date, in Bayesian approach there is no literature on local sensitivity analysis in the presence of general missingness of outcome and covariates. In this article, I extend the ISNI (index of sensitivity to local nonignorability) approach of [Zhang and Heitjan \(2007\)](#), [Gao et al. \(2016\)](#) and [Momeni Roochi and Eftekhari Mahabadi \(2024\)](#) in two directions: 1) I allow general missingness in outcomes and covariates which is not necessarily simultaneous. 2) I use a Bayesian approach that incorporates prior information.

In Section 2, I derive the Bayesian sensitivity indices. In Section 3, I conduct a simulation study to evaluate the performance of these indices. Section 4 offers further discussion.

2 Bayesian Local Sensitivity

This section develops the Bayesian ISNI, for general missingness in response and covariates. It presents the log-likelihood function, log-posterior function and proposes the Bayesian local sensitivity indices.

2.1 The log-likelihood and log-posterior function

The dataset comprises N independent vectors (Y_i, X_i, Z_i) , wherein Y_i represents an outcome that is subject to missingness, while X_i is a covariate also afflicted by missingness, and Z_i denotes a covariate vector that is devoid of any missingness. The variables R_i and S_i serve as indicators of missingness: $R_i = 1(0)$ if Y_i is observed (missing); $S_i = 1(0)$ if X_i is observed (missing). We can readily extend the definitions of R_i and S_i to vector forms in the context of multivariate Y_i and X_i . For simplicity we factor the conditional distribution of (R_i, S_i, Y_i, X_i) given Z_i as $f_{\xi, \gamma, \theta_y, \theta_x}^{R_i, S_i, Y_i, X_i | Z_i}(r_i, s_i, y_i, x_i | z_i) = f_{\xi}^{R_i | S_i, Y_i, X_i, Z_i}(r_i | s_i, y_i, x_i, z_i) \times f_{\gamma}^{S_i | Y_i, X_i, Z_i}(s_i | y_i, x_i, z_i) \times f_{\theta_y}^{Y_i | X_i, Z_i}(y_i | x_i, z_i) \times f_{\theta_x}^{X_i | Z_i}(x_i | z_i)$ and henceafter write $f_{\theta_y}^{Y_i | X_i, Z_i}(y_i | x_i, z_i)$, $f_{\theta_x}^{X_i | Z_i}$ as f_{θ_y} , f_{θ_x} , respectively. In addition, I denote the probabilities that Y_i and X_i are observed as, respectively,

$$f_{\xi}^{R_i | S_i, Y_i, X_i, Z_i}(1 | s_i, y_i, x_i, z_i) = \begin{cases} h(q_{\xi}(\cdot) + \xi_y y_i) & \text{if } s = 1 \\ h(q_{\xi}(\cdot) + \xi_x x_i + \xi_y y_i) & \text{if } s = 0 \end{cases} \quad (2.1)$$

$$f_{\gamma}^{S_i|Y_i,X_i,Z_i}(1 | y_i, x_i, z_i) = \begin{cases} h(q_{\gamma}(\cdot) + \gamma_y y_i) & \text{if } r = 1 \\ h(q_{\gamma}(\cdot) + \gamma_x x_i + \gamma_y y_i) & \text{if } r = 0 \end{cases} \quad (2.2)$$

where $h(\cdot)$ is the inverse of an appropriate link function and $q_{\xi}(\cdot)$ and $q_{\gamma}(\cdot)$ are identifiable functions of observed values. In addition, $\boldsymbol{\nu} = (\xi_y, \xi_x, \gamma_x, \gamma_y)^T$ is the vector of non-ignorability parameters. The assumption $\boldsymbol{\nu} = \mathbf{0}$ leads to the MAR missingness mechanism, which results in the ignorable model. The log-likelihood is then

$$\begin{aligned} l = & \sum_{i:g_i=1, s_i=1} \left\{ \ln(h(q_{\xi}(\cdot) + \xi_y y_i)) + \ln(h(q_{\gamma}(\cdot) + \gamma_x x_i)) + \ln f_{\boldsymbol{\theta}_y}(y_i | x_i, z_i) + \ln f_{\boldsymbol{\theta}_x}(x_i | z_i) \right\} \\ & + \sum_{i:g_i=0, s_i=1} \ln \left\{ \int \left(1 - h(q_{\xi}(\cdot) + \xi_y y_i^{miss}) \right) \times h(q_{\gamma}(\cdot) + \gamma_x x_i + \gamma_y y_i^{miss}) \right. \\ & \quad \left. \times f_{\boldsymbol{\theta}_y}(y_i^{miss} | x_i, z_i) \times f_{\boldsymbol{\theta}_x}(x_i | z_i) dy_i^{miss} \right\} \\ & + \sum_{i:g_i=1, s_i=0} \ln \left\{ \int h(q_{\xi}(\cdot) + \xi_x x_i^{miss} + \xi_y y_i) \times (1 - h(q_{\gamma}(\cdot) + \gamma_y y_i)) \right. \\ & \quad \left. \times f_{\boldsymbol{\theta}_y}(y_i | x_i^{miss}, z_i) \times f_{\boldsymbol{\theta}_x}(x_i^{miss} | z_i) dx_i^{miss} \right\} \\ & + \sum_{i:g_i=0, s_i=0} \ln \left\{ \int \int (1 - h(q_{\xi}(\cdot) + \xi_x x_i^{miss} + \xi_y y_i^{miss})) \times (1 - h(q_{\gamma}(\cdot) + \gamma_x x_i^{miss} + \gamma_y y_i^{miss})) \right. \\ & \quad \left. \times f_{\boldsymbol{\theta}_y}(y_i^{miss} | x_i^{miss}, z_i) \times f_{\boldsymbol{\theta}_x}(x_i^{miss} | z_i) dx_i^{miss} dy_i^{miss} \right\} \end{aligned} \quad (2.3)$$

The parameters of main interest $\boldsymbol{\theta}_y$ index the conditional distribution of the outcome Y given covariates (X, Z) . The rest are nuisance parameters: $\boldsymbol{\theta}_x$ govern the distribution of X given Z ; $(q_{\gamma}, \gamma_x, \gamma_y)$ governs the missingness mechanism of S given (Y, X, Z) ; and (q_{ξ}, ξ_x, ξ_y) governs the missingness mechanism of R given (S, Y, X, Z) . Let $\mathbf{q} = (q_{\xi}, q_{\gamma})$. To conduct Bayesian inference, the joint prior density function $\pi(\boldsymbol{\theta}_y, \boldsymbol{\theta}_x, \mathbf{q}) = \pi(\boldsymbol{\theta}_y, \boldsymbol{\theta}_x)\pi(\mathbf{q})$, is assigned to the parameters of the response, missing covariate and ignorable missing mechanism models. Implementing the Bayes theorem, the log-posterior function of $(\boldsymbol{\theta}_y, \boldsymbol{\theta}_x, \mathbf{q})$ is

$$\ln \pi(\boldsymbol{\theta}_y, \boldsymbol{\theta}_x, \mathbf{q} | y, x, z, r, s; \boldsymbol{\nu}) \propto \ln \pi(\boldsymbol{\theta}_y, \boldsymbol{\theta}_x) + \ln \pi(\mathbf{q}) + l(\boldsymbol{\theta}_y, \boldsymbol{\theta}_x, \mathbf{q} | y, x, z, r, s; \boldsymbol{\nu}), \quad (2.4)$$

where $l(\cdot)$ is the log-likelihood function of $(\boldsymbol{\theta}_y, \boldsymbol{\theta}_x, \mathbf{q})$ in a selection model, given in equation (2.3). The above posterior distribution does not possess a closed-form solution due to the integrals with respect to the non-ignorability parameters; therefore, numerical methods are required. When $\boldsymbol{\nu} = \mathbf{0}$, equation (2.4) simplifies to the ignorable posterior density function. Conversely, when

using a non-ignorable Bayesian model for data analysis, the posterior density function becomes complex due to the computation of integrals involving missing response and covariates. This leads to time-consuming calculations. This method is particularly useful when the assumption of ignorability is questioned and a non-ignorable Bayesian model is used for data analysis.

2.2 Bayesian index of sensitivity to non-ignorability

Zhang and Heitjan (2007), Eftekhari Mahabadi (2018) and Momeni Roochi and Eftekhari Mahabadi (2024) advocated for the execution of a sensitivity analysis by evaluating the variation of the Bayes estimator of θ_y as a function of the nonignorability parameter in the neighbourhood of the missing at random model. I extend this framework to encompass scenarios in which both dependent variables and independent variables may be missing by employing a first-order Taylor expansion of the Bayes estimator of θ_y under squared error loss (S.E.L) given a fixed value of ν (i.e. $\hat{\theta}_{yB}(\nu)$) at $\nu = \mathbf{0}$:

$$\hat{\theta}_{yB}(\nu) \simeq \hat{\theta}_{yB}(\mathbf{0}) + \nu^T \frac{\partial \hat{\theta}_{yB}(\nu)}{\partial \nu} \Big|_{\nu=\mathbf{0}}, \quad (2.5)$$

where $\hat{\theta}_{yB}(\mathbf{0})$ is the ignorable Bayes estimator of θ_y . Local sensitivity analysis considers whether small values of the nonignorability parameter ν can yield nonnegligible changes in the Bayes estimates. Following Zhang and Heitjan (2007), I define the index of local sensitivity to nonignorability as $\Lambda^1(\theta_y) = \frac{\partial \hat{\theta}_{yB}(\nu)}{\partial \nu} \Big|_{\nu=\mathbf{0}}$. Then a general formula for $\Lambda^1(\theta_y)$ is

$$\Lambda^1(\theta_y) = \begin{pmatrix} Cov_I(\theta_y, \frac{\partial l(\theta_y, \theta_x, \mathbf{q}|y, x, z, r, s; \nu)}{\partial \xi_y} \Big|_{\nu=\mathbf{0}}) \\ Cov_I(\theta_y, \frac{\partial l(\theta_y, \theta_x, \mathbf{q}|y, x, z, r, s; \nu)}{\partial \xi_x} \Big|_{\nu=\mathbf{0}}) \\ Cov_I(\theta_y, \frac{\partial l(\theta_y, \theta_x, \mathbf{q}|y, x, z, r, s; \nu)}{\partial \gamma_x} \Big|_{\nu=\mathbf{0}}) \\ Cov_I(\theta_y, \frac{\partial l(\theta_y, \theta_x, \mathbf{q}|y, x, z, r, s; \nu)}{\partial \gamma_y} \Big|_{\nu=\mathbf{0}}) \end{pmatrix}, \quad (2.6)$$

Each element of the $\Lambda^1(\theta_y)$ vector in (2.6) is the posterior covariance between the parameters θ_y and the derivative of the non-ignorable log-likelihood with respect to a non-ignorability parameter, evaluated at $\nu = \mathbf{0}$. This covariance is typically intractable but can be approximated numerically.

3 Simulation Study

I apply ISNI to some simulated datasets from the Bayesian simple linear regression model with the possibility of general missing outcome and covariate.

Without any loss of generality, I will consider a linear regression model that includes a single normally distributed covariate with the possibility of general missingness with the response variable.

Therefore, Y_i and X_i follow a bivariate normal distribution, represented as $(Y_i, X_i)^T \sim N_2(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, with the mean vector $\boldsymbol{\mu} = (\beta_0 + \beta_1\mu_x, \mu_x)^T$ and the covariance matrix $\boldsymbol{\Sigma} = \begin{pmatrix} \sigma_y^2 & \beta_1\sigma_x^2 \\ \beta_1\sigma_x^2 & \sigma_x^2 \end{pmatrix}$, where Y_i and X_i are the outcome and covariate for the i th subject, respectively. β_0 and β_1 represent regression coefficients, whereas the marginal variance of Y is calculated as $\sigma_y^2 = \beta_1^2\sigma_x^2 + \sigma_{y|x}^2$, where $\sigma_{y|x}^2$ and σ_x^2 denote the variances of the conditional distribution of Y given X , and the marginal distribution of X , respectively. Some random samples, each with a size of $N = 500$ are generated. The random samples are simulated based on the aforementioned simple linear regression model with the set of true parameter values $\beta_0 = 1$, $\beta_1 = 2$, $\sigma_{y|x}^2 = \sigma_x^2 = 1$ and $\mu_x = 0$. Additionally, missing values in both the outcome and covariate are generated using equations (2.1) and (2.2) with a logit link function. For simplicity, I consider $q_\xi(\cdot) = \alpha_{01y}$ and $q_\xi(\cdot) = \alpha_{02y}$ for the first and second equations of (2.1), respectively. Furthermore, $q_\gamma(\cdot) = \alpha_{0x}$ in (2.2). To conduct Bayesian inference, the Jeffreys' prior for $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and non-informative prior $N(0, 10^3)$ is considered for α_{01y} , α_{02y} and α_{0x} . The posterior samples are obtained via Independent Metropolis-Hastings algorithm with multivariate multivariate normal proposal distribution. The location parameter of this proposal distribution is assumed to be the ML estimate of parameters β_0 , β_1 , $\sigma_{y|x}^2$, σ_x^2 , α_{0x} , α_{01y} and α_{02y} , and the scale parameter is the inverse of the negative Hessian matrix. I generated 50000 MCMC samples along with 20000 burn-ins. A range of different missing rates is obtained by specifying distinct values for α_{01y} , α_{02y} and α_{0x} within the missingness mechanisms (2.1) and (2.2). These values are set as follows: 1) $\alpha_{01y} = 1$, $\alpha_{02y} = -0.9$, $\alpha_{0x} = -0.5$, 2) identical to setting 1, with the exception of $\alpha_{01y} = 0.5$, 3) identical to setting 1, with the exception of $\alpha_{01y} = 1.5$, 4) identical to setting 1, with the exception of $\alpha_{0x} = 0.5$, 5) identical to setting 1, with the exception of $\alpha_{0x} = 1$, 6) identical to setting 1, with the exception of $\alpha_{0x} = 1.5$, 7) identical to setting 1, with the exception of $\alpha_{0x} = 2$, 8) identical to setting 1, with the exception of $\alpha_{0x} = 2.5$. In addition, the true values of the vector of non-ignorability parameters, ν , are set as $(-0.3, 1, -0.8, 2)$. To obtain reliable results, each missingness occurrence scenario was repeated 100 times. In each repetition, the values of $\boldsymbol{\Lambda}^1(\boldsymbol{\theta}_y)$ in (2.6) are calculated for parameters of β_0 and β_1 . Subsequently, the approximate Bayesian estimates for these parameters are calculated using the first-order Taylor expansion (2.5). For comparison, Bayesian estimates via complete-case analysis and an ignorable model are also obtained. Table 1 presents the average missing rates in the covariate (Xrate) and response (Yrate), along with the absolute biases of the Bayesian estimates under complete-case analysis, the ignorable model (ign) and the first-order Taylor expansion approximation (appr) of $\boldsymbol{\theta}_y = (\beta_0, \beta_1)$. The posterior

standard errors of Bayesian estimates (sd) are also shown. According to these tables, the biases of the Bayesian estimates obtained by the first-order Taylor expansion approximation are, on average, clearly smaller than those of the other methods, indicating the superiority of ISNI. To

Table 1: Average results of Bayesian parameter estimation and approximate estimates via first-order Taylor expansion in Bayesian local sensitivity analysis of β_0 and β_1 for eight simulated datasets with $N = 500$ individuals, in different missing rates

Scenario	Missing Rate		β_0					β_1				
	Xrate	Yrate	Bias.cc	sd.cc	Bias.ign	sd.ign	Bias.appr	Bias.cc	sd.cc	Bias.ign	sd.ign	Bias.appr
1	0.376	0.382	0.451	0.089	0.427	0.101	0.354	0.549	0.096	0.249	0.092	0.042
2	0.404	0.418	0.575	0.094	0.479	0.122	0.040	0.425	0.091	0.250	0.092	0.062
3	0.432	0.338	0.404	0.080	0.285	0.087	0.197	0.596	0.089	0.145	0.082	0.006
4	0.282	0.328	0.432	0.068	0.201	0.077	0.305	0.568	0.078	0.149	0.075	0.009
5	0.248	0.312	0.279	0.070	0.250	0.072	0.199	0.721	0.073	0.117	0.067	0.065
6	0.252	0.168	0.276	0.064	0.317	0.068	0.046	0.724	0.069	0.196	0.065	0.060
7	0.230	0.192	0.193	0.055	0.245	0.057	0.030	0.807	0.061	0.108	0.057	0.101
8	0.206	0.162	0.282	0.057	0.203	0.062	0.132	0.718	0.066	0.100	0.063	0.071

explore these superiority, figure 1 shows the boxplot of the biases obtained in eight scenarios for each method and parameter. This figure indicates that the biases of first-order Taylor expansion approximation are smaller than other methods.

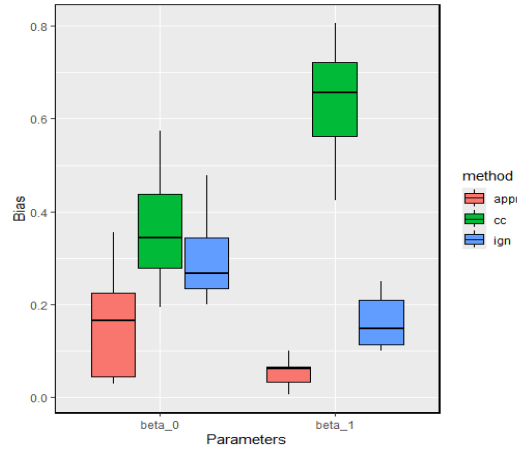


Figure 1: Comparison of absolute values of biases of Bayesian estimates in complete-case (cc) and ignorable (ign) analysis with the first-order Taylor expansion approximation (appr) of β_0 and β_1 in eight simulated datasets.

4 Discussion and Conclusions

This paper has introduced novel Bayesian local sensitivity indices to address the critical, yet under-studied, problem of non-ignorable general missingness, where the response variable and covariates are missing that is not necessarily simultaneous. The proposed indices, $\mathbf{\Lambda}^1(\theta_y)$ efficiently quantify the sensitivity of posterior estimates locally around the more tractable ignorable (MAR) model. This provides a powerful diagnostic tool for a complex missing data problem without requiring the fitting of a full non-ignorable model.

My extensive simulation studies, designed specifically around scenarios of general missingness in simple linear regression, demonstrate that the Bayesian estimates via ISNI are more accurate than other methods like complete-case analysis and ignorable model.

An important limitation of the current work, as with most local sensitivity approaches, is its reliance on the correct parametric specification of the selection model. A promising extension to mitigate this limitation would be to develop my index within a more flexible, semiparametric Bayesian selection model, building on recent advances in this area [Sugasawa et al. \(2022\)](#). This would enhance the robustness of the sensitivity analysis to parametric assumptions.

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